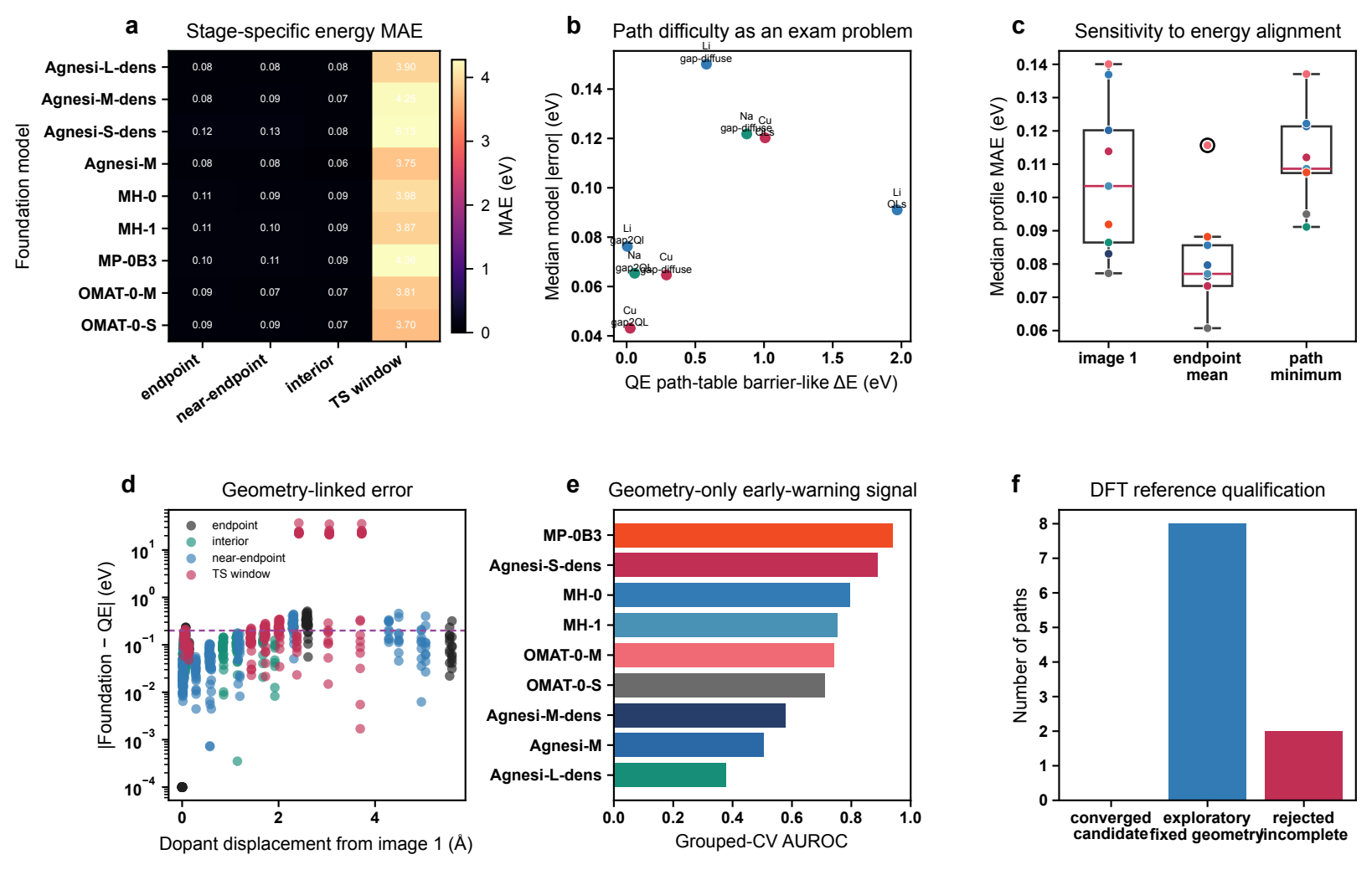

a
Stage-specific energy MAE
Foundation model
Agnesi-L-dens
Agnesi-M-dens
Agnesi-S-dens
Agnesi-M
MH-0
MH-1
MP-0B3
OMAT-0-M
OMAT-0-S
0.08 0.08 0.08 3.90
0.08 0.09 0.07 4.25
0.12 0.13 0.08 6.13
0.08 0.08 0.06 3.75
0.11 0.09 0.09 3.98
0.11 0.10 0.09 3.87
0.10 0.11 0.09 4.36
0.09 0.07 0.07 3.81
0.09 0.09 0.07 3.70
endpoint
near-endpoint
interior
TS window
MAE (eV)
0
1
2
3
4
b
Path difficulty as an exam problem
Median model |error| (eV)
QE path-table barrier-like ΔE (eV)
Li gap-diffuse
Na gap-diffuse
Cu gap-diffuse
Li QLs
Li gap2QL
Na gap2QL
Cu gap-diffuse
Cu gap2QL
c
Sensitivity to energy alignment
Median profile MAE (eV)
image 1
endpoint mean
path minimum
d
Geometry-linked error
|Foundation − QE| (eV)
endpoint
interior
near-endpoint
TS window
Dopant displacement from image 1 (Å)
e
Geometry-only early-warning signal
MP-0B3
Agnesi-S-dens
MH-0
MH-1
OMAT-0-M
OMAT-0-S
Agnesi-M-dens
Agnesi-M
Agnesi-L-dens
Grouped-CV AUROC
f
DFT reference qualification
Number of paths
converged candidate
exploratory fixed geometry
rejected incomplete